Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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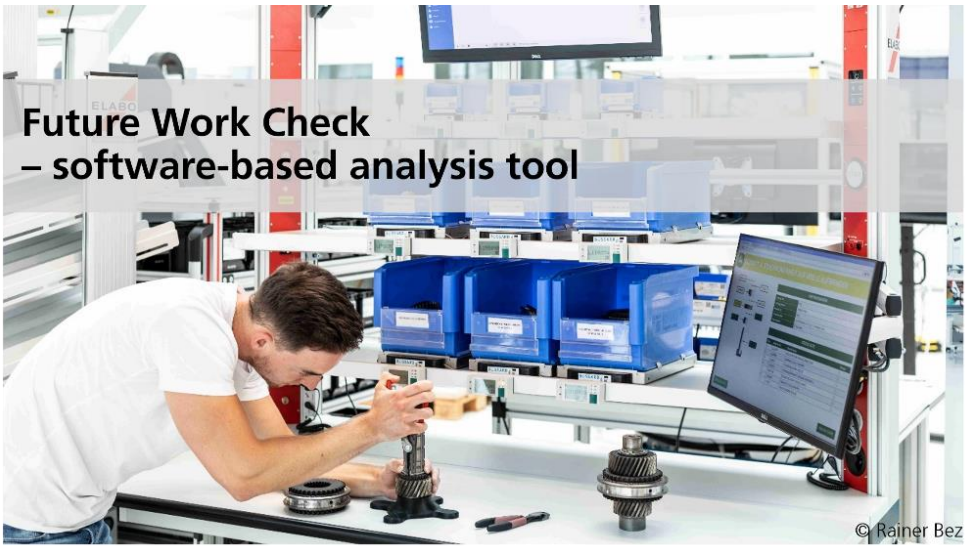
2. Technology Overview

Title of Proposed Technology	**Future Work Check – software-based analysis tool** Our software-based analysis method enables industrial engineers to optimize workplaces in industrial production by mastering complexity and generating precise calls for action. Time savings: from 6+ days down to ~1 day for collecting data and generating workplace improvement ideas. - High-quality recommendations: precise actions on how to improve workplaces. - Emotional benefit: a "carefree" feeling by ensuring all relevant aspects are covered. - Happy stakeholders: through a multi-perspective approach.  Figure 1: Future Work Check – software-based analysis tool (© Rainer Bez, Fraunhofer IPA)

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Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	*Production Management / Quantitative Analysis Techniques*	*Software*	*Software Solutions*

Concept of Proposed Technology

The Future Work Check is a digital method and toolchain for the holistic analysis, representation, and evaluation of industrial work systems. It integrates socio-technical systems theory with a software-based assessment approach, enabling systematic and efficient workplace analysis beyond traditional paper- or spreadsheet-based audits. The concept is structured based on two logical layers complementing each other (see Figure 2, Hall et al. 2024).

First, the **theoretical foundation** provides a validated socio-technical framework that ensures a holistic perspective on manufacturing work systems by systematically addressing human, technological, and organizational dimensions and their interactions. This foundation defines the assessment criteria, maturity logic, and evaluation mechanisms (Hall et al. 2024).

Second, the **application level** translates this foundation into a modular digital toolchain that is designed to be easily and barrier-free accessible for small and medium-sized enterprises. It comprises (1) structured data collection via a user-friendly web application, (2) a digital representation of each analyzed work system using interoperable data structures, and (3) automated evaluation logic that derives maturity scores, work system profiles, and tailored calls to action (Hall et al. 2024).

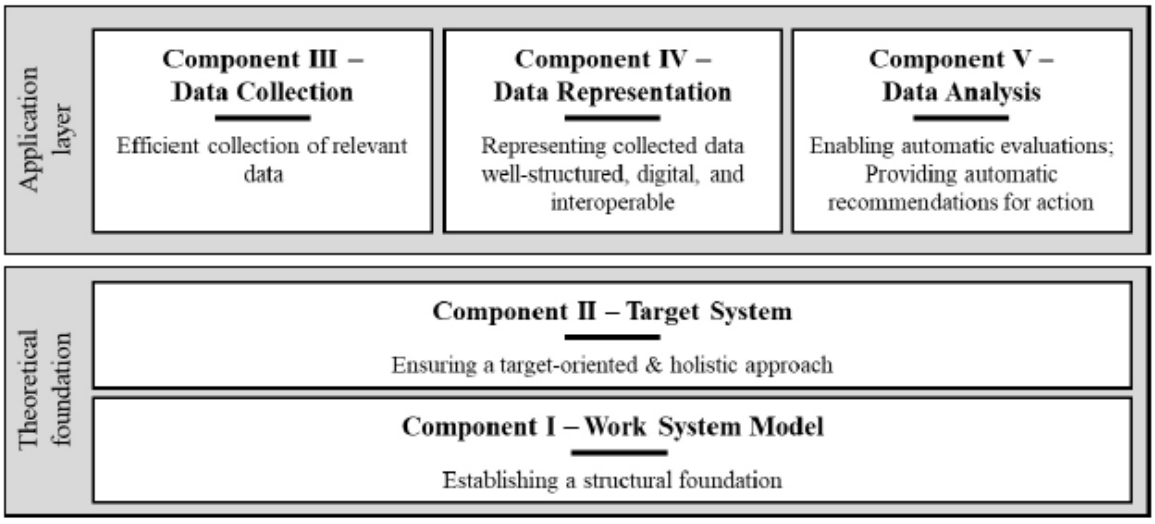

Figure 2: Scientific concept of the Future Work Check (Hall et al. 2024)

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	The outcomes of this novel approach cover a broad spectrum of improvement measures, including ergonomics, age-appropriate workplace design, lean production, digitalization, and the application of artificial intelligence. They deliberately combine immediately implementable quick wins with more comprehensive, project-oriented improvement initiatives, enabling both short-term impact and longer-term strategic development. By combining methodological rigor with digital automation, the Future Work Check becomes a navigation system that enables industrial engineers and SMEs to analyze complex work systems in a consistent, scalable, and resource-efficient manner, and to derive actionable guidance for the systematic development of industrial workplaces.
Rationale for Joint R&D Project	Manufacturing SMEs, globally as well as especially in Korea, are facing **rising system complexity driven by digital transformation** (Kwak et al 2023), **increasing product variety** (Lee et al. 2016), and **demographic change** (Jones 2025). At the same time, workplace and work system redesign is often slowed down by fragmented, non-digital assessment practices that focus on isolated aspects (e.g., ergonomics or technology only) and lack a holistic, systematic evaluation logic. As a result, improvement potential remains largely unused, redesign decisions are insufficiently evidence-based, and change initiatives become costly, inconsistent, and difficult to sustain (Hall et al 2024). Addressing this challenge requires an integrated, digitally supported and socio-technical approach that is independent of existing IT infrastructures and can be applied with minimal organizational and technical effort. A joint R&D project is essential because successful deployment and commercialization in Korea depend on (1) systematic adaptation of the method to Korean manufacturing contexts and SME-specific constraints, (2) validation of usability and effectiveness under real SME, and (3) refinement of the approach through application in multiple Korean pilot sites. The collaboration builds on complementary strengths. **Fraunhofer IPA** contributes a scientifically validated socio-technical method, maturity-based evaluation logic, and the digital toolchain concept. The **Korean SME partner** contributes in-depth knowledge of local manufacturing practices, access to representative pilot environments, and market-specific requirements for adoption and scaling. Only through this joint R&D approach can the Future Work Check be transformed from a validated method into a market-ready, scalable solution tailored to the needs of Korean manufacturing SMEs.
Key Words	1. Process optimization 2. Industrial work systems 3. Socio-technical systems 4. Artificial intelligence 5. Digital twin

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Prototype Stage (5-6)	End of the Project	Productization Stage (7-8)
R&D Objectives	**Objective O1 – Digital Work System Representation (Interoperable)** Develop a structured and interoperable digital representation of industrial work systems ("Work System Twin") based on modular information models. The representation enables reuse, benchmarking, and cross-case analysis of work systems while deliberately avoiding dependencies on existing shopfloor or enterprise IT systems. **O2 – Automated and AI-Supported Maturity-Based Evaluation Logic** Develop an automated evaluation logic based on maturity levels and target-oriented weighting of objectives, complemented by AI-supported pattern recognition to identify typical maturity profiles, development pathways, and improvement priorities across comparable work systems. **O3 – AI-Supported Calls-to-Action Generation and Prioritization** Develop an AI-supported logic for generating and prioritizing structured, context-sensitive calls to action derived from evaluation results. The calls to action cover ergonomics, age-appropriate workplace design, lean production, digitalization, and AI use cases, and explicitly combine immediately implementable quick wins with more comprehensive, project-oriented improvement initiatives. Trade-offs between competing objectives (e.g., productivity, flexibility, autonomy, well-being) are handled transparently to support informed human decision-making. **O4 – Rapid, Low-Barrier Data Collection and Intelligent Consistency Support** Design and validate a guided, web-based assessment workflow that enables fast and low-effort data collection, reuse of existing documentation, and completion of comprehensive work system analyses within hours rather than days. AI-based assistance supports consistency checks, plausibility validation, and user guidance without increasing data collection complexity. **O5 – Adaptation, Validation, and Responsible AI Use in Korean Manufacturing SMEs** Adapt the method and toolchain to Korean manufacturing contexts and SME-specific requirements, and validate usability, effectiveness, and AI-supported functionalities across multiple Korean pilot work systems. Particular emphasis is placed on transparency, explainability, and responsible use of AI in decision-support for industrial workplace design.			
R&D Target Specifications	**Target T1 – Time and cost savings (→ Objectives 2, 3, 4):** Reduce the end-to-end assessment time per work system (data collection, evaluation, and generation of calls to action) from typically 4–5 working days to ≤ 1 working day. **T2 – On-site data collection effort (→ O4):** Limit structured on-site data collection effort to ≤ 3 hours per work system as a design target, excluding preparation and follow-up activities. **T3 – Low-barrier applicability (→ O4):** Enable full application of the method without technical integration into existing MES, ERP, or shopfloor IT systems, requiring only standard web access and basic workplace documentation. **T4 – Interoperability at data-model level (→ O1):** Support standardized export and import of structured work-system data using open, documented data formats (e.g., JSON-based models aligned with modular information structures), enabling reuse, comparison, and benchmarking. **T5 – Calls-to-action coverage and quality (→ O3):** For each evaluation criterion and maturity level, provide at least one clearly formulated call to action, including: (a) classification as quick win or project-oriented measure; (b) transparent indication of relevant trade-offs where applicable (e.g., productivity vs. autonomy). **T6 – AI-supported prioritization and consistency (→ O2, O3):** Demonstrate AI-supported prioritization of calls to action and automated consistency checks that reduce manual evaluation effort by ≥ 30 % compared to a purely manual baseline, while maintaining transparent and explainable outputs.			

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	T7 – Benchmarking capability (→ O1, O2): Enable benchmarking of work systems against reference profiles and peer work systems within the Future Work Check dataset (work system vs. reference; work system vs. work system). **T8 – Usability and acceptance (→ O5):** Achieve a System Usability Scale (SUS) score ≥ 75 among industrial users in pilot applications. **T9 – Pilot validation scope (→ O5):** Validate the method and toolchain in at least 10 work systems across a minimum of 2 Korean manufacturing SMEs, demonstrating measurable reductions in assessment effort and improved decision quality.
R&D Plan	**Phase I (Pre-study, Months 1-6): "Korea Fit & Feasibility"** Short pre-study phase to validate requirements, adapt the Future Work Check concept to the Korean SME context, and define a proof-of-concept and development roadmap for the main R&D phase. **WP1 – Requirements Analysis of Korean Manufacturing SMEs** Objective addressed: O1, O5 • Analyze Korean SMEs manufacturing specifics, constraints, and decision-making practices • Identify priority work system types (e.g., assembly, machining, logistics, maintenance) • Derive functional and non-functional requirements for the Future Work Check (e.g., usability, assessment duration, transparency, low-barrier applicability) • Identify key evaluation dimensions and target KPIs relevant for Korean SMEs Outputs: • Korean SME requirements specification • Prioritized use cases and work system typology **WP2 – Adaptation of the Future Work Check Concept** Objective addressed: O1, O2, O5 • Adapt the Future Work Check framework, criteria set, and maturity logic to SME requirements • Translate model contents from German to sufficient language (English, Korean) • Adapt calls-to-action structure and content to Korean manufacturing practices • Define the conceptual design of the lightweight digital toolchain, including digital work system representation and AI-supported evaluation principles • Ensure consistency with low-barrier applicability and responsible, explainable AI use Outputs: • Adapted FWC concept and criteria blueprint • Translated model available in sufficient language (English, Korean) for application **WP3 – Proof of Concept and Phase II Roadmapping** Objective addressed: O1, O2, O3, O4, O5 • Define and implement a limited proof of concept to demonstrate core principles (e.g., assessment flow, evaluation logic, calls-to-action generation) • Validate conceptual feasibility with selected Korean SME stakeholders • Define the detailed development roadmap, milestones, and go/no-go criteria for Phase II Outputs: • Proof-of-concept demonstration • Phase II development roadmap and milestones
	Phase II (Main R&D project, Months 7-36): "Develop, Apply, Validate" Development of the digital and AI-based core components of the Future Work Check, their integration into a usable toolchain, and validation in Korean manufacturing SMEs. **WP4 – Developing the Digital Work System Twin** Objective addressed: O1

- Design and implement an interoperable digital representation of industrial work systems
- Structure work system data along human, technological, and organizational dimensions
- Enable comparison, and reuse of work system representations
- Support benchmarking across work systems and reference profiles

Outputs:

- Implemented Digital Work System Twin
- Benchmarking-ready work system data model

WP5 – Developing the AI-Based Evaluation and Decision-Support Methodology

Objective addressed: O2, O3

- Develop automated, maturity-based evaluation logic for work system assessment
- Implement AI-supported pattern recognition across assessed work systems
- Develop AI-supported prioritization and consistency checks for evaluation results
- Generate structured, explainable calls to action covering ergonomics, age-appropriate workplace design, lean production, digitalization, and AI use cases

Outputs:

- AI-supported evaluation and decision-support methodology
- Explainable calls-to-action logic

WP6 – Toolchain Development for Practical Application

Objective addressed**:** O1, O4

- Integrate digital work system representation and AI-based evaluation into coherent toolchain
- Implement guided data collection workflows and role-based user perspectives
- Enable reuse of existing workplace documentation; ensure low-barrier usability

Outputs:

- Integrated Future Work Check toolchain
- Usability-validated assessment workflow

WP7 – Application, Pilot Validation, and Iterative Refinement

Objective addressed**:** O4, O5

- Apply the toolchain in multiple work systems within Korean manufacturing SMEs
- Conduct pilot studies and collect quantitative and qualitative feedback
- Validate achievement of SMART R&D target specifications

Outputs**:**

- Validated pilot results and KPI evidence
- Refined Future Work Check (Korea edition)

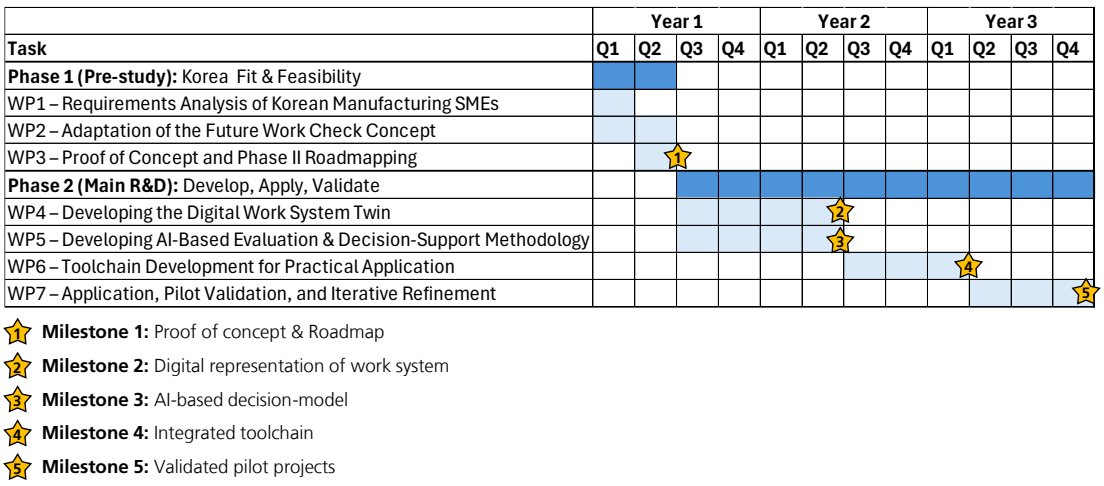

Figure 3: Project plan and milestones

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IPA will act as the **scientific lead and methodological backbone** of the project, providing interdisciplinary research expertise, conceptual guidance, and advanced software prototyping capabilities. Fraunhofer IPA provides qualified **research staff** with expertise in industrial engineering, software development, and applied research. The research team is professionally structured according to a **3H role model (Hipster – Hustler – Hacker)** to ensure effective translation of research into validated, industry-relevant solutions. **Technical Research and Software Prototyping (Hacker Role)** • Design and implement **research-oriented software prototypes** and proof-of-concept systems. • Develop and maintain the **core technical architecture** of the research prototype. • Integrate novel algorithms, data models, or system components derived from research outcomes. • Support testing, debugging, and iterative refinement of the prototype with the SME. • Ensure code quality, documentation, and reproducibility for research and technology transfer. **Strategic and Conceptual Leadership (Hipster Role)** • Lead the **conceptual design and scientific framing** of the project, including research questions, hypotheses, and evaluation criteria. • Translate industrial challenges into **research-driven solution concepts**. • Ensure **scientific rigor**, novelty, and alignment with international research standards. • Supervise the overall research direction and ensure coherence across work packages. • Contribute to dissemination, including **scientific publications, reports, and presentations**. **Project Coordination and Research-to-Application Translation (Hustler Role)** • Act as the **project coordination interface** between research and industry partners. • Structure and manage work packages, milestones, and deliverables from research perspective. • Ensure the **transfer of research results into applicable prototypes and demonstrators**. • Take responsibility for **requirements engineering**, prioritization, and roadmap definition in close collaboration with the Korean SME. • Provide continuity and ownership of the solution, incl. alignment with exploitation perspectives.
Expected R&R of Korean SME Partner	The Korean SME partner will primarily contribute **practical domain knowledge and application-oriented requirements** derived from its real working environment. The SME's main responsibility is to provide insights into existing workplaces, processes, and operational constraints, thereby enabling the research team to ground the development in real-world conditions. The SME will benefit from the identification and assessment of potential improvement measures that can be derived from the Future Work Check and adapted for later implementation. At a minimum, the SME will appoint a **dedicated contact person** who serves as the main interface to the research institute and facilitates access to relevant processes, workplaces, and contextual information. This person will be supported by the research team throughout the project. Optionally, and depending on availability and relevance, the SME may involve additional internal stakeholders such as industrial engineering, occupational safety, supervisors, or employees. Collaboration activities will mainly take place through **structured workshops and interviews**, complemented where necessary by the provision of existing documentation for detailed analysis.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological outcomes** A validated software toolchain for fast, holistic work system assessment and improvement planning, including interoperable Digital Work System Twin representation and automatic scoring/ recommendations. Expected technological outcomes are: 1) **Digital Work System Twin** 2) **AI-supported evaluation** and **calls-to-action methodology** 3) Integrated, low-barrier **Future Work Check toolchain** 4) Validated **application results** from Korean SMEs
	Commercialization potential After successful completion of the R&D project, the software solution shows strong commercialization potential as a **scalable product tailored to the needs of Korean SMEs**. It enables companies to significantly reduce the effort associated with workplace redesign and continuous improvement by providing a structured, data-driven assessment approach. The solution supports **consistent benchmarking across multiple work systems and sites**, allowing SMEs to identify improvement potentials in a comparable and repeatable manner. To ensure broad market adoption, the software is designed with **deployment options aligned with typical SME constraints**, including easy-to-use data collection processes and the reuse of existing documentation, minimizing implementation effort while maximizing practical value.
	Industrial and economic impact After successful completion of the R&D project, the software solution is expected to generate **significant industrial and economic impact at scale**, particularly for manufacturing-oriented SMEs. By reducing the engineering effort required per workplace assessment from several days to **less than one day**, companies can transition from isolated, selective analyses to **systematic, portfolio-wide assessments** across entire plants or multiple sites. On an industrial scale, this efficiency gain enables organizations to assess **dozens or even hundreds of workplaces annually**, translating into substantial cumulative cost savings and freeing engineering capacity for higher-value activities such as optimization and implementation rather than analysis. Beyond direct efficiency gains, the solution accelerates the **early identification of improvement measures** related to productivity, quality, and human-centered work design. At an economic level, this supports shorter improvement cycles, more consistent decision-making, and reduced risk of suboptimal or delayed interventions. Over time, such effects can contribute to **measurable productivity improvements, reduced quality losses, and improved workforce sustainability**, strengthening the competitiveness of SMEs and supporting broader industrial resilience in high-wage manufacturing environments.

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☐ Terms:

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1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization